Law of Supply

General Economics

Supply

Willing to Offer to the Market at Various Prices during Period of Time

Able to Offer to the Market at Various Prices during Period of Time

Supply

What Firms Offer for Sale, Not Necessarily to What they Succeed in Selling

Is a Flow i.e. as per unit of time, per day, per week, or per year

Definitions of Supply

- **The Supply of Goods is the Quantity offered for Sale in a given Market at a given Time at various Prices.**

By : Thomas

- **Supply refers to the Amounts of a Good that Producer in a given Market Desire to Sell, during a given Time Period at Various Prices, Ceteris Paribus.**

By : Samuelson

Determinants of Supply

- Price of the Good
- Price of Related Goods
- Price of the Factors of Production
- State of Technology
- Government Policy
- Other Factors

Determinants of Supply

- **Price of the Commodity**

Ceteris Paribus i.e. Other Things Being Equal,

Relative Price of the Good ↑

Quantity Supplied ↑

This Happens Because Goods are Produced by the Firm to Gain Profits. Profit rises when Price rises.

Determinants of Supply

- Price of the Related Good

Price of Related Good (Y) ↑

Quantity Supplied of Other Good (X) ↓

Rise in Price of Related Good makes it more Profitable for the Firm to Produce & Sell.

Determinants of Supply

- **Prices of the Factors of Production**

Change in Price of Factors of Production

Changes in Relative Profitability of Different Lines of Production

Producers Shift from one Line to Another

Supplies of Different Commodities Change

Determinants of Supply

- Government Policy
  - Imposition of Commodities Taxes Increase the Cost of Production.
  - Subsidies Reduce the Cost of Production which Increases Firm's Supply.

Determinants of Supply

- State of Technology
- Other Factors
  - Govt. Industrial & Foreign Policies
  - Goals of the Firm
  - Market Structure, etc.

Law of Supply

- **Law of Supply states that other things being equal, the Higher the Price, the Greater the Quantity Supplied or the Lower the Price, the Smaller the Quantity Supplied.**

By : Dooley

- **The Law of Supply states that Other things being Equal, the Quantities of any Commodity that Firms will Produce & Offer for Sale, is Positively related to the Commodities own Price, Rising when Price Rises & Falling when Price Falls.**

By : Lipsey

Law of Supply

- There is a Direct Relationship Between Price & Quantity Supplied:
  - Quantity Supplied Rises as Price Rises, Other things Constant.
  - Quantity Supplied Falls as Price Falls, Other things Constant.
- The Law of Supply is accounted for by 2 Factors:
  - When Prices Rise, Firms Substitute Production of One Good for Another.
  - Assuming firms' Costs are Constant, a Higher Price means Higher Profits.

Law of Supply

- Behaviour of Supply Depends upon:
  - Phenomenon Considered.
  - Degree of Possible Adjustment in Supply.
  - Time taken into Consideration i.e. Short-Run & Long Run.

Assumption to Law of Supply

- **Law of Supply holds Good when "Other Things Remain the Same" meaning thereby, the Factors affecting Supply ,other than Price, are Assumed to be Constant.**

- **Supply Function: $Q_x = f(P_X, C_x, T_x)$**

  where, Q_x = Supply of Commodity X

  P_x = Price of Commodity X

  C_x = Cost of Production of Commodity X

  T_x = Technology of its Production

Supply Schedule

- Supply Schedule is a Series of Quantities which Producer would like to Sell per unit of Time at Different Prices.
- Two Aspects of Supply Schedule
  - Individual Supply Schedule
  - Market Supply Schedule

Individual Supply Schedule

- It is defined as a Table which shows Quantities of a Given Commodity which an Individual Producer will Sell at all Possible Prices at a given Time.

Price (Rs.) (per kg)	Quantity Supplied (kg)
1	10
2	30
3	50
4	70
5	80

Market Supply Schedule

- **It is defined as the Quantities of a Given Commodity which all Producers will Sell at all Possible Prices at a given Moment of Time. In Market there are many Producers of a Single Commodity. By Aggregating the Individual Supply, the Market Supply Schedule is Constructed.**

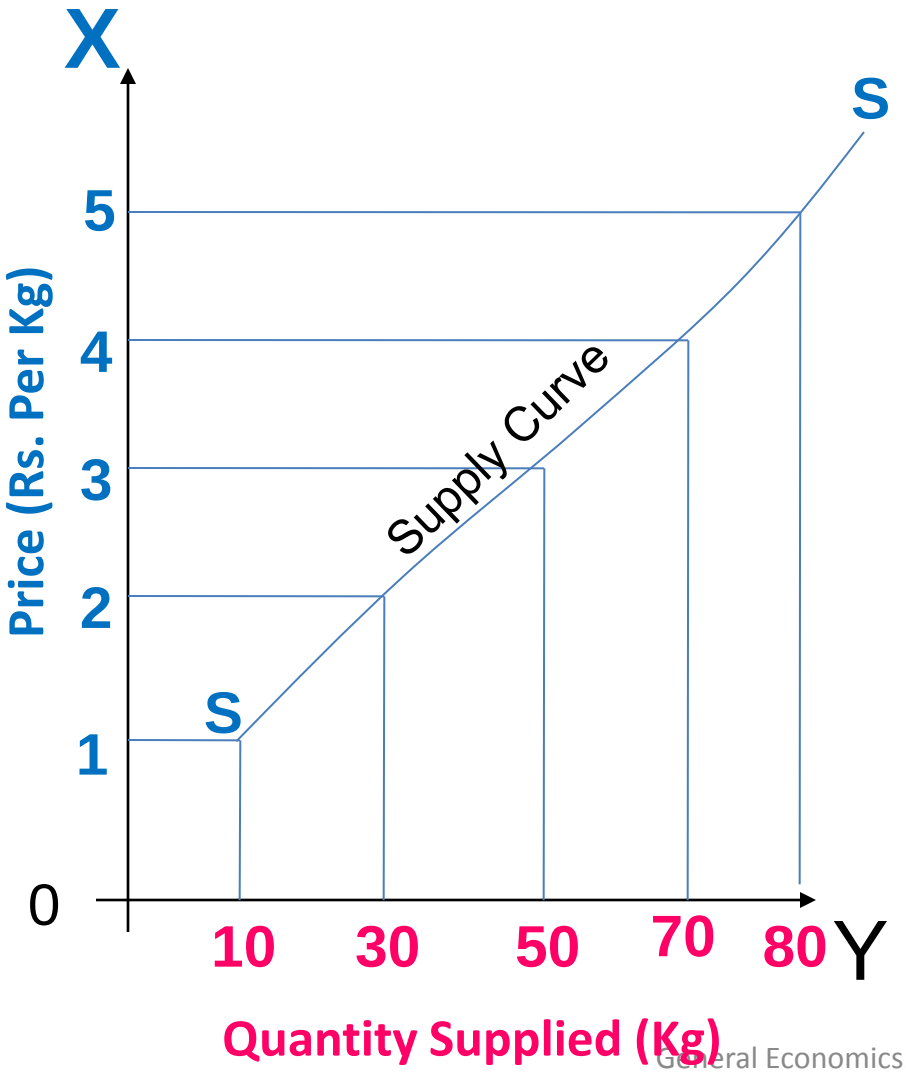
Price of Commodity 'X' (in Rs.)	Supply by A	Supply by B	Market Supply (Units)
100	40	50	40+50=90
200	60	70	60+70=130
300	65	80	65+80=145
400	80	100	80+100=180

It indicates that when Price of 'X' is Rs 100 per unit, A's Supply is of 40 units and that of 'B' is of 50 units. Thus the Market Supply is 90 units. As the Price Increases, Quantity Supplied Increases.

Supply Curve

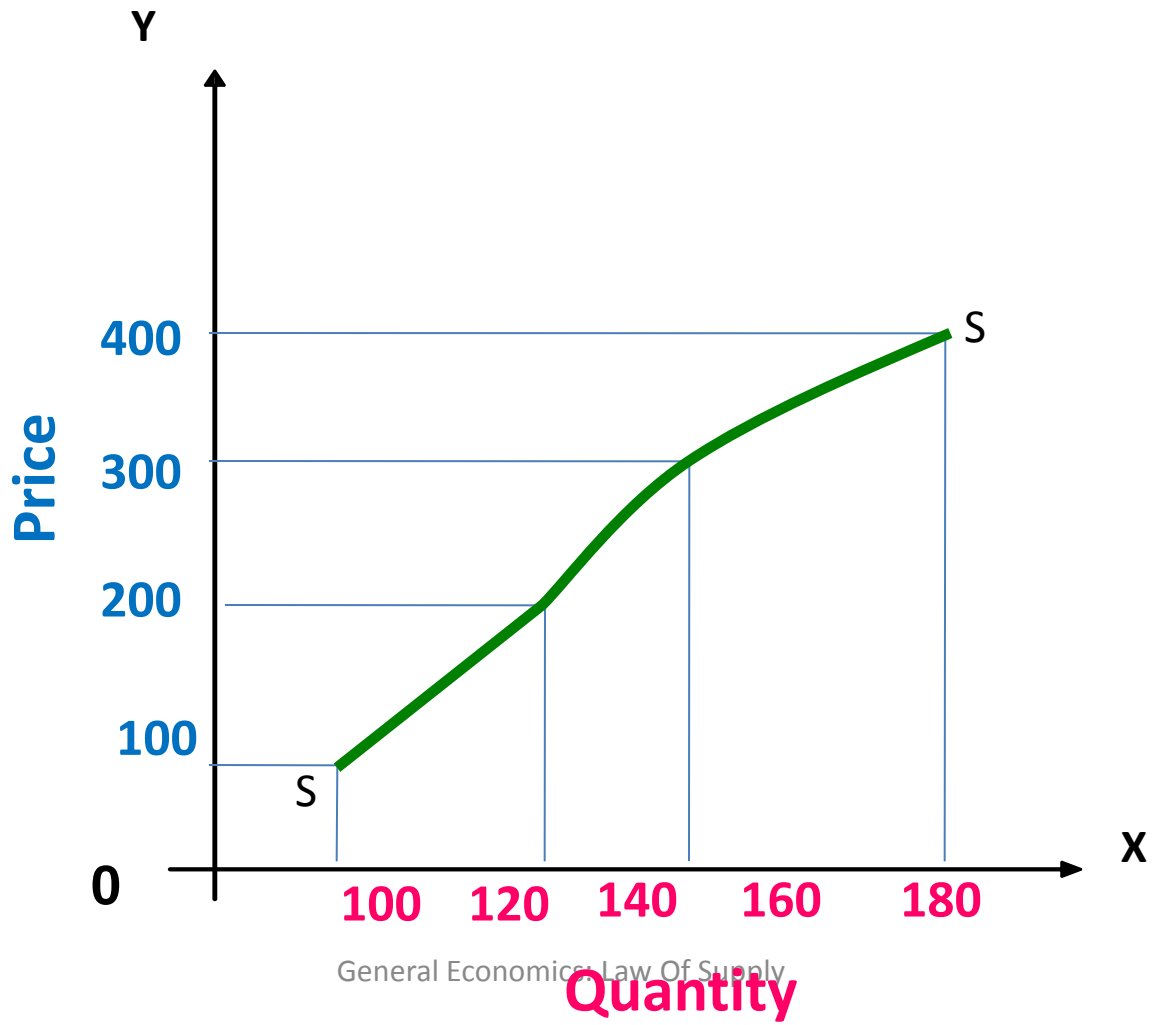
- A Supply Curve is a Locus of Points showing various Price-Quantity Combinations of a Seller.
- It shows the Direct Relationship between Price & Quantity Supplied.
- It Slopes Upwards to the Right.

Individual Supply Curve



The Supply Curve Slopes Upwards from Left to Right, meaning thereby that when Price is High Quantity Supplied is also High and vice versa.

Market Supply Curve



Exceptions to Law of Supply

- **Supply of Labour: If we take the Supply of Labour at very High Wages, we may find that the Supply of Labour has decreased instead of Increasing.**
- **Agricultural Products: Since the Production of Agricultural Products cannot be Increased beyond a certain Limit, the Supply cannot be Increased beyond this Limit even on an Increase in their Prices.**

Exceptions to Law of Supply

- **Artistic Goods** : **Supply of Artistic Goods cannot be Increased or Decreased easily.**
- **Goods of Auction**: **Supply of Goods of Auction is Limited as such cannot neither be Increased nor Decreased.**
- **Hope of Change in the Prices of Commodities in Near Future**: **If the Price of Commodity is on Rising Pace, then the Supply of such Commodity Decreases as Producers and Sellers will like to Store this Commodity & Vice-Versa.**

Expansion & Contraction in Supply

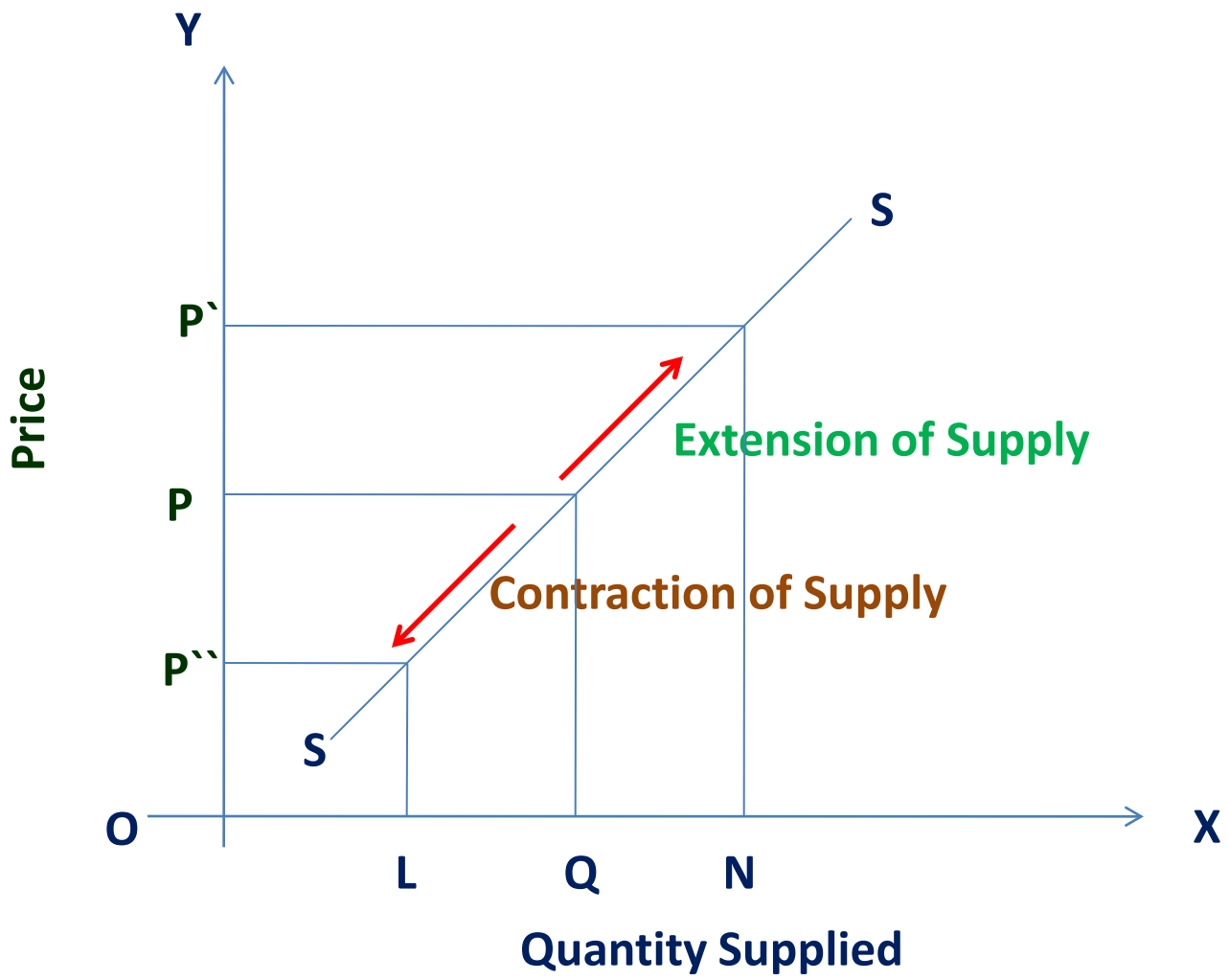
Expansion

- QS ↑ Price ↑
- Upward Movement Along the Supply Curve

Contraction

- QD ↓ Price ↓
- Downward Movement Along the Supply Curve

Extension & Contraction in Supply



Increase & Decrease in Supply

Increase

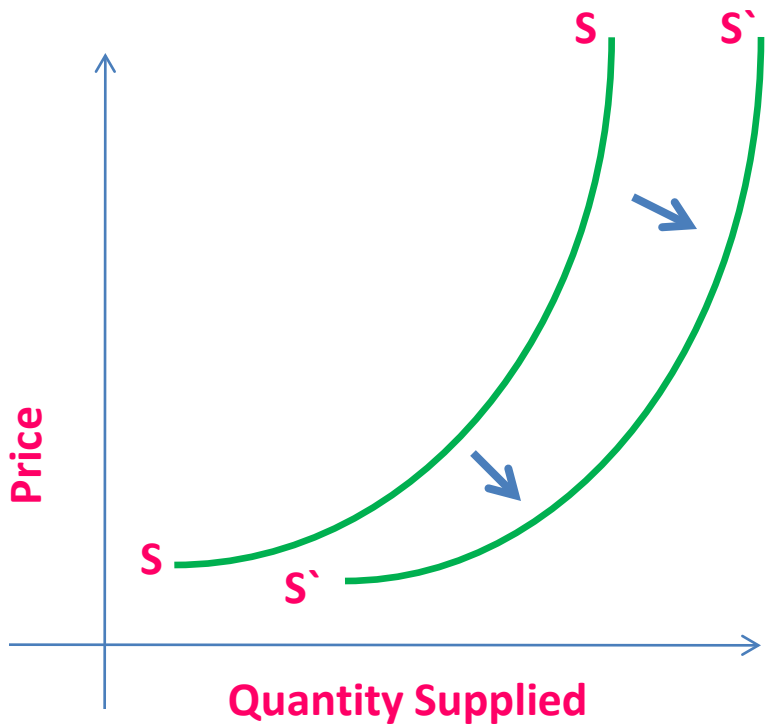
- Q Supplied ↑ (at all prices) due to Change in Other Factors
- Rightward Shift

Decrease

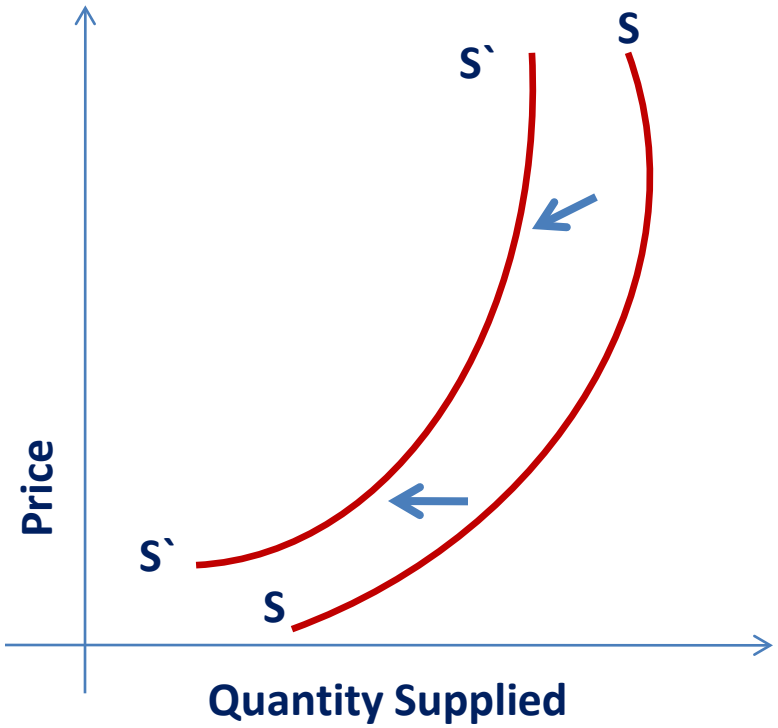
- Q Supplied ↓ (at all prices) due to Change in Other Factors
- Leftward Shift

Increase & Decrease in Supply

Increase in Supply



Decrease in Supply



Elasticity of Supply

- **Elasticity of Supply is defined as the Responsiveness of the Quantity Supplied of a Good to Change in its Price.**

$$E_S = \frac{\%\ \text{Change in Q. Supplied}}{\%\ \text{Change in Price}}$$

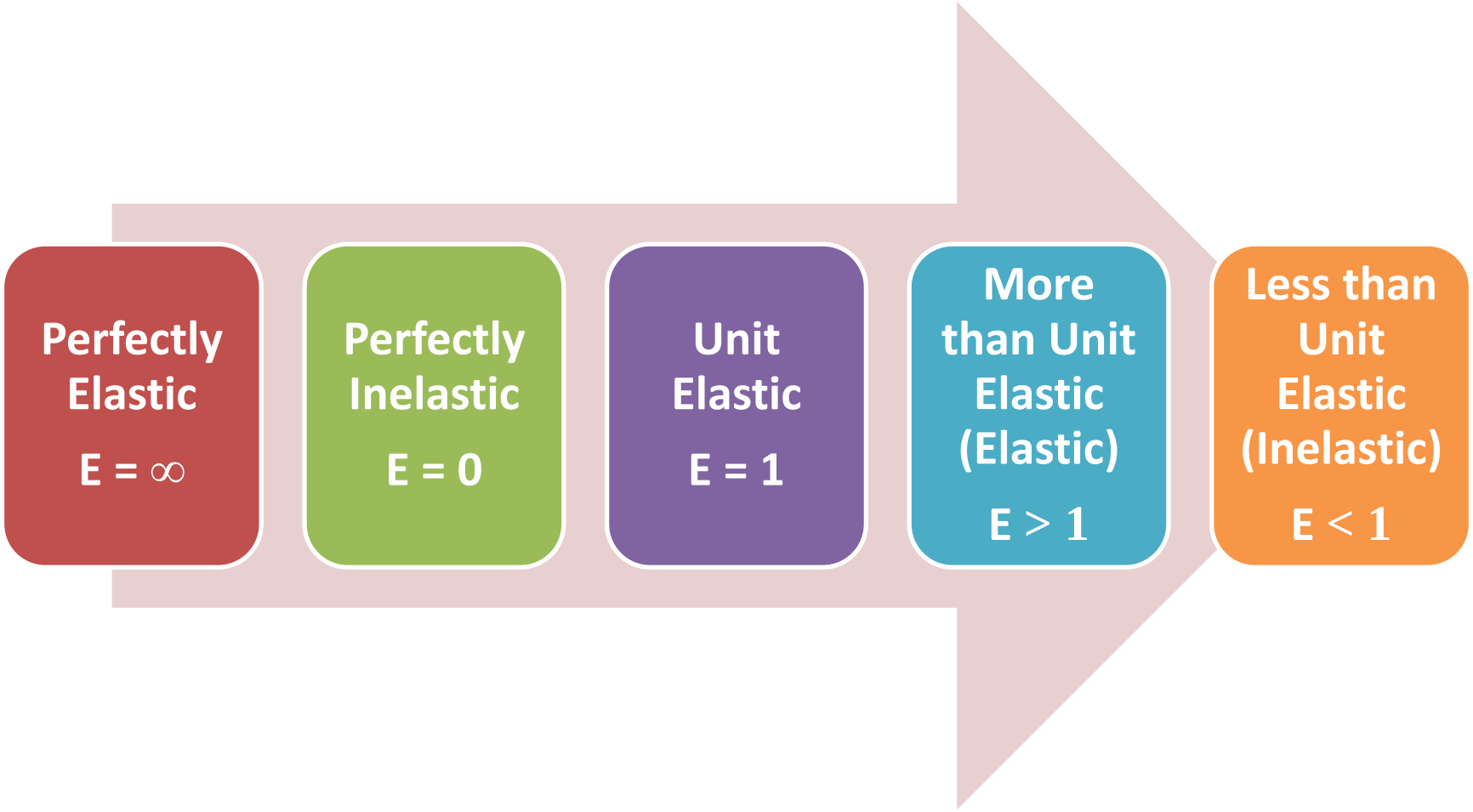
$$E_S = \frac{\text{Change in Q. Supplied}}{\text{Change in Price}} \times \frac{\text{Original Price}}{\text{Q. Supplied}}$$

Elasticity of Supply

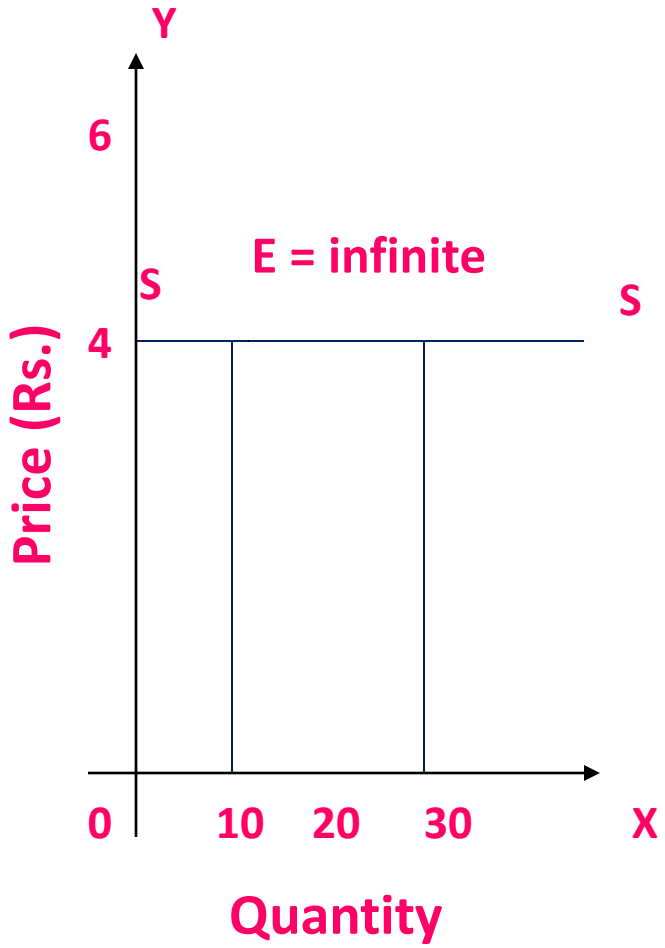
$$E_S = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

Where,	E_S	**Price Elasticity of Supply**
	ΔQ	**Change in Quantity Supplied**
	Q	**Original Quantity Supplied**
	ΔP	**Change in Price**
	P	**Original Price**

Degrees of Price Elasticity of Supply

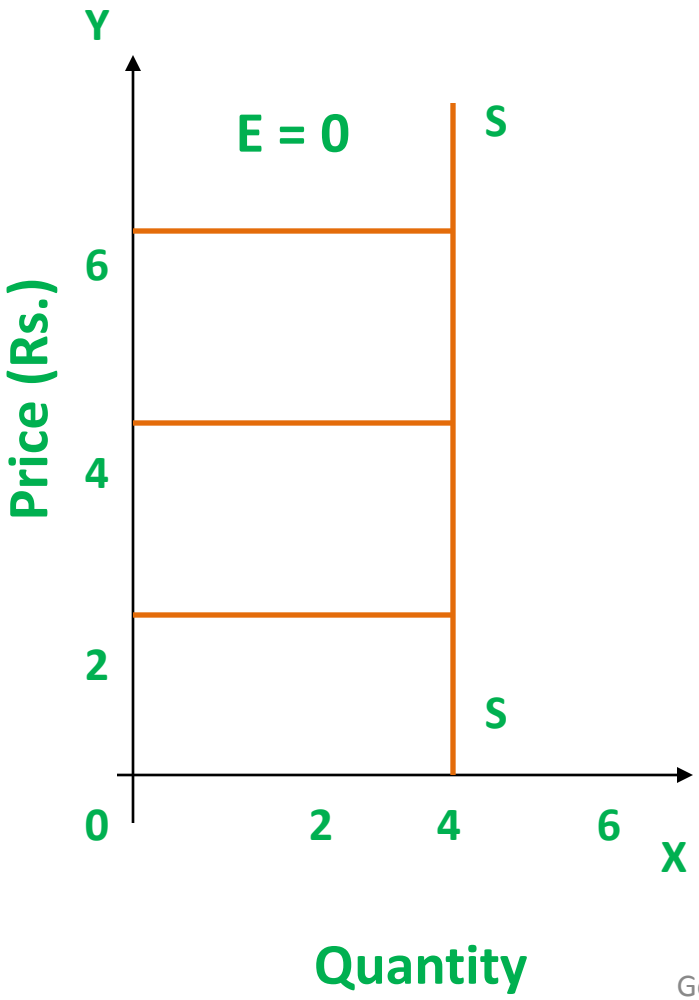


Perfectly Elastic Supply



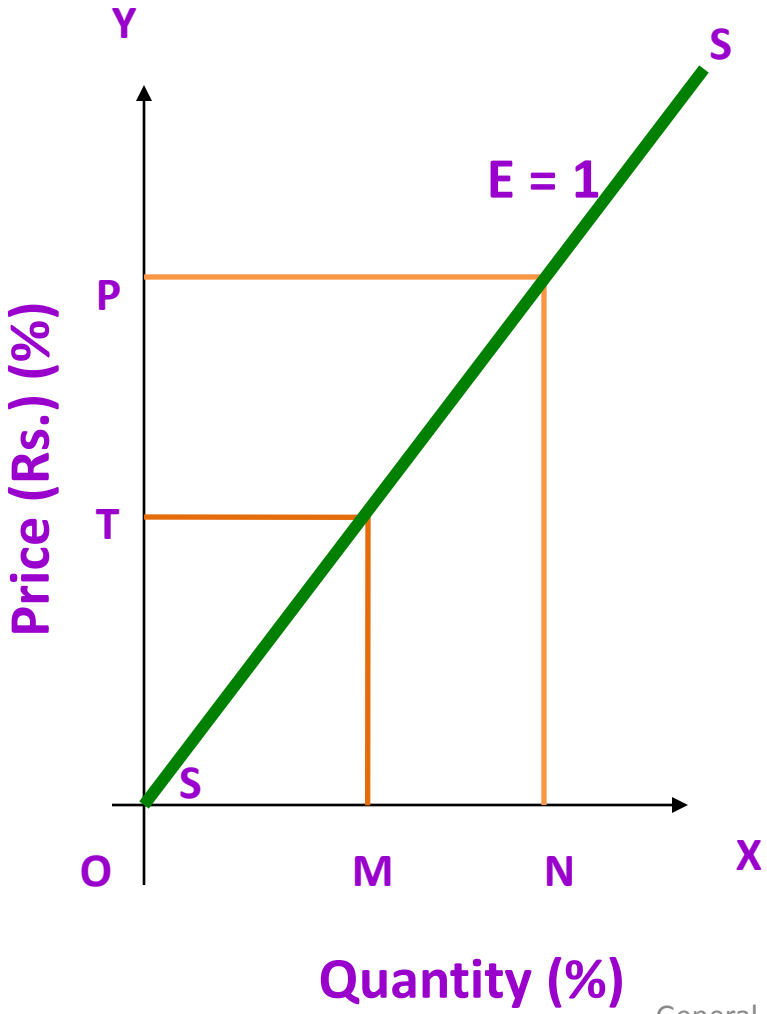
- **A Perfectly Elastic Supply is one in which there is a Significant Change in the Supply of the Commodity without any Change or Little Change in its price.**
- **It is an Imaginary Concept. In Practical Life, there is no Commodity, the Supply of which is Perfectly Elastic.**

Perfectly Inelastic Supply



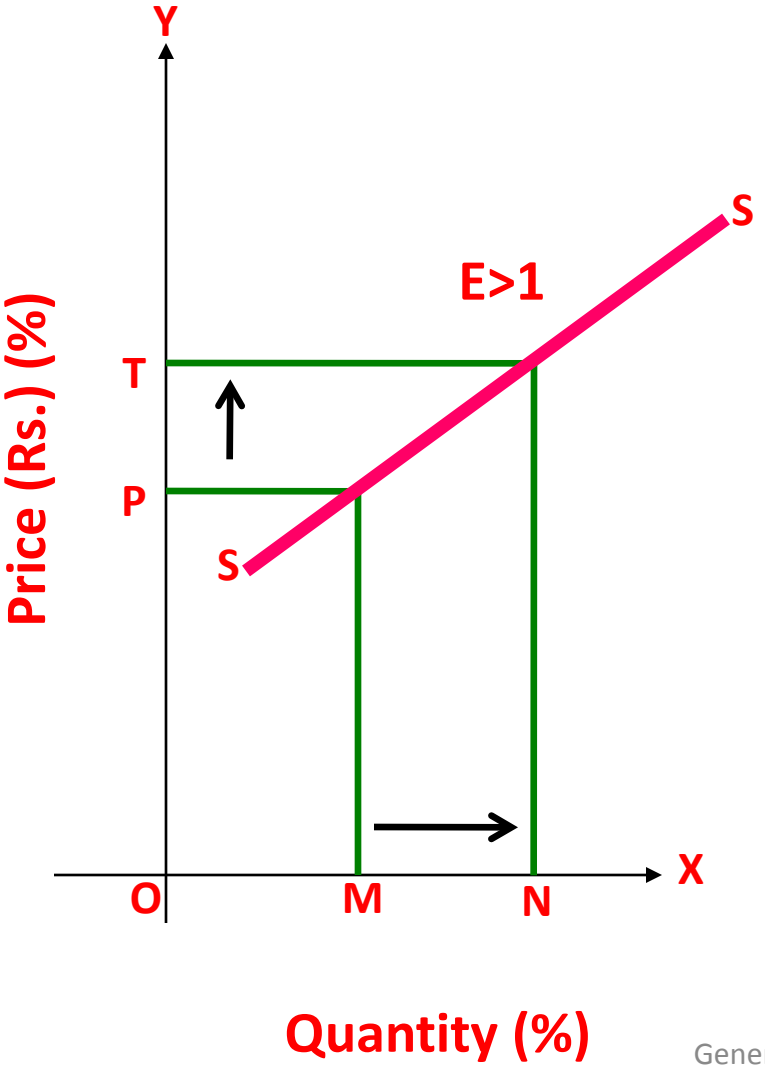
- Perfectly Inelastic Supply is one in which a Change in Price Produces No Change in the Quantity Supplied.
- It is an Imaginary Concept. In Practical Life, there is no Commodity, the Supply of which is Perfectly Inelastic.

Unitary Elastic Supply



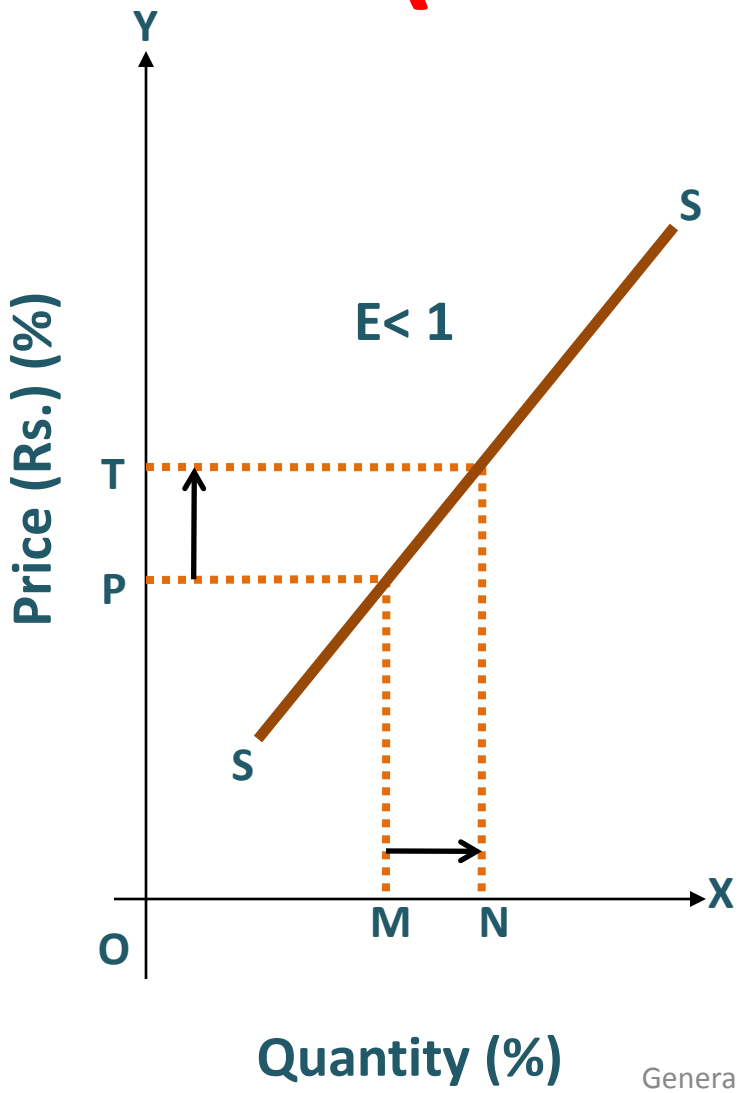
- **Unitary Elastic Demand is one in which a % Change in Price Produces an Equal % Change in Quantity Supplied.**

Greater than Unitary Elastic (Elastic) Supply



- Greater than Unitary Elastic Supply is one in which a Given % Change in Price Produces Relatively more % Change in Supply.

Less than Unitary Elastic (Inelastic) Supply



- Less than Unitary Elastic Demand is one in which a given % Change in Price Produces Relatively Less % Change in Quantity Supplied.

Point Elasticity of Supply

- Refers to Measuring the Elasticity at a Particular Point on Supply Curve.
- Makes Use of Derivative Changes Rather than Finite Changes in Price & Quantity Supplied.
- Defined As:

$$\frac{dq}{dp} \times \frac{p}{q}$$

Where, $\frac{dq}{dp}$ is the Differentiation of Supply Function w.r.t. Price at a point on Supply Curve.

Arc Elasticity of Supply

- **When Elasticity is to be found between 2 Points, we use Arc Elasticity.**

$$\text{Elasticity} = \frac{q_1 - q_2}{q_1 + q_2} \times \frac{p_1 + p_2}{p_1 - p_2}$$

Where,

p_1 = Original Price

q_1 = Original Quantity Supplied

p_2 = New Price

q_2 = New Quantity Supplied

Arc Elasticity of Supply

For Example, Find Elasticity of Supply Between:

p_1 = Rs. 12 $q_1 = 20$

p_2 = Rs. 15 $q_2 = 50$

$$\text{Elasticity} = \frac{q_1 - q_2}{q_1 + q_2} \times \frac{p_1 + p_2}{p_1 - p_2}$$

$$E_s = \frac{30}{70} \times \frac{27}{3}$$

$$E_s = +3.85$$

Determinants of Price Elasticity of Supply

- **Nature of Commodity:**

Perishable → • Inelastic Supply

Durable → • Elastic Supply

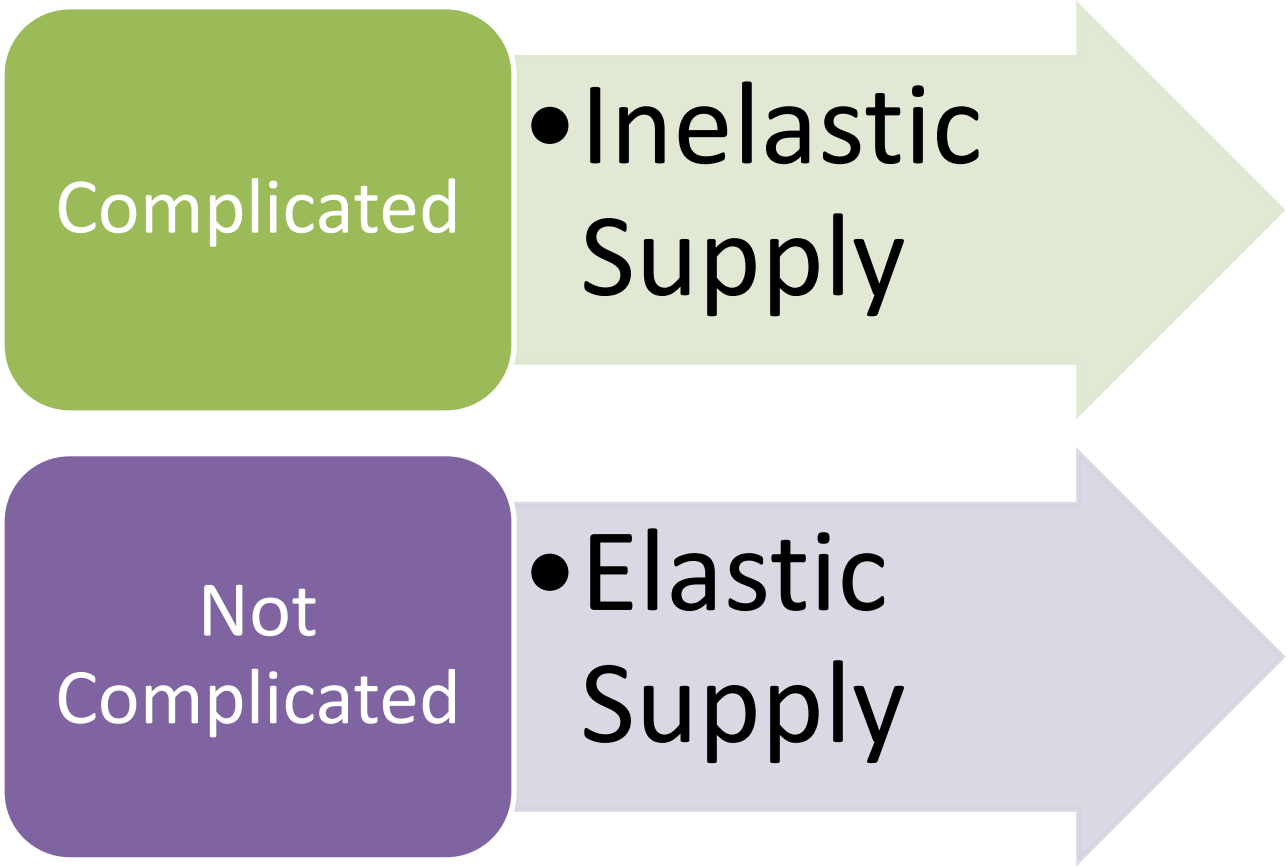
Determinants of Price Elasticity of Supply

- **Time**

Very Short Period	• Inelastic
Short Period	• Elastic
Long Period	• Highly Elastic

Determinants of Price Elasticity of Supply

- **Production Technique**



Determinants of Price Elasticity of Supply

- **Stages of Law of Returns**

Law of Diminishing Returns	• Inelastic
Law of Constant Returns	• Elastic
Law of Increasing Returns	• Highly Elastic

Q 1

The Supply of a Good refers to;

a) **Actual Production of a Good**

b) **Total Existing Stock of a Good**

c) **Stock available for Sale**

d) **Amount of a Good offered for Sale at a particular Price per unit of Time**

Q 2

A Vertical Supply Curve parallel to Y Axis implies that the Elasticity of Supply is:

a) Zero

b) Infinity

c) Equal to One

d) Greater than Zero but less than Infinity

Q 3

An Increase in the Supply of a Good is caused by:

a) **Improvements in its Technology**

b) **Fall in the Price of other Goods**

c) **Fall in the Prices of Factors of Production**

d) **All of the above**

Q 4

Elasticity of Supply refers to the degree of responsiveness of Supply of a Good to changes in its:

a) **Demand**

b) **Price**

c) **Cost of Production**

d) **State of Technology**

Q 5

A Horizontal Supply Curve parallel to Quantity Axis implies that the Elasticity of Supply is:

a) Zero

b) Infinity

c) Equal to One

d) Greater than Zero but less than One

Q 6

Contraction of Supply is the result of:

a) Decrease in the number of producers

b) Decrease in the Prices of the Goods concerned

c) Increase in the Prices of other Goods

d) Decrease in the outlay of Sellers

Q 7

Supply of a Commodity is a:

a) Stock Concept

b) Flow Concept

c) Both Stock and Flow Concept

d) None of these

Q 8

If the Price of apple rises from Rs. 30 per Kg to Rs. 40 per Kg and the Supply increases from 240 Kg to 300 Kg. Elasticity of Supply is:

a) 0.77

b) 0.67

c) (-) 0.67

d) (-) 0.77

Q 9

Contraction of Supply is the result of:

a) Decrease in the number of Producers

b) Decrease in the Price of Good concerned

c) Decrease in the Price of other Goods

d) None of the above

Q 10

When Quantity Supplied changes by larger percentage than does Price, Elasticity is termed as:

a) Inelastic

b) Perfectly Elastic

c) Elastic

d) Perfectly Inelastic

Q 11

If the Elasticity of Supply is Zero then Supply Curve will be:

a) **Horizontal**

b) **Downward Sloping**

c) **Upward Sloping to the right**

d) **Vertical**

Q 12

If as a result of change in Price the Quantity Supplied of a Good remains unchanged, we conclude that:

a) **Elasticity of Supply is Perfectly Inelastic**

b) **Elasticity of Supply is Relatively Greater Elastic**

c) **Elasticity of Supply is Inelastic**

d) **Elasticity of Supply is Relatively Less Elastic**

Q 13

Period in which Supply cannot be increased Is called:

a) **Market Period**

b) **Short Run**

c) **Long Run**

d) **None of These**

Q 14

Supply of Good and its Price have:

a) Negative Relationship

b) Inverse Relationship

c) No Relationship

d) Positive Relationship

Q 15

An Expansion in the Supply of Good is caused by a:

a) **Rise in the Price of Good**

b) **Fall in the Prices of Other Goods**

c) **Fall in the Prices of Factors of Production**

d) **All of the Above**

Q 16

Which of the following have the Lowest Price Elasticity of Supply?

a) **Luxury**

b) **Necessities**

c) **Salt**

d) **Perishable Goods**

Q 17

Which of the following Method is not used for Measuring Elasticity of Supply?

a) Arc Method

b) Percentage Method

c) Total Outlay Method

d) Point Method

Q 18

Other Things Remaining Constant, the Law of Supply States:

a) **Supply of Commodities is Directly related to its Price**

b) **Price is not related to Supply**

c) **As Supply Rises, Price also Rises**

d) **Supply is not related to Factors Other than Supply**

Q 19

Generally Supply Curve of Industrial Products is

a) **Positively Sloped**

b) **Negatively Sloped**

c) **Both (a) And (b)**

d) **Parallel to Y-Axis**

Q 20

Elasticity of Durable Goods is:

a) Perfectly Inelastic

b) Unitary Elastic

c) Elastic

d) Inelastic

Q 21

All of the Following are Determinants of Supply Except

a) **Prices of Factors of Production**

b) **State of Technology**

c) **Income of Consumer**

d) **Price of Related Goods**

Q 22

The Exception to the Law of Supply are are:

a) Artistic Goods

b) Auction Goods

c) Agricultural Products

d) All of the Above

Q 23

Supply Curve in most cases Slopes

a) Upward towards Right

b) Vertical And Parallel to Y-axis

c) Upward Towards Left

d) Horizontal And Parallel to X-axis

Q 24

Behaviour of Supply depends upon:

a) Time Taken into Consideration

b) Degree of Possible Adjustment in Supply

c) Both (a) & (b)

d) Only (b)

Q 25

Leftward Shift of the Supply Curve Refers to:

a) Expansion in Supply

b) Increase in Supply

c) Contraction in Supply

d) Decrease in Supply

THE END

Law of Supply